

Md. Jahidul Islam Sarker

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Professional Summary

Computer Science undergraduate with strong foundations in algorithms, systems programming, and full-stack development. Built production-style applications using React and Spring Boot, and solved 700+ competitive programming problems across Codeforces, CodeChef, LeetCode, and CSES. Experienced in REST APIs, real-time systems with WebSockets, and JWT-based authentication. Actively participates in CTF competitions, with top rankings in web exploitation, cryptography, and reverse engineering. Interested in backend engineering, distributed systems, and cybersecurity.

Skills

- **Languages:** C, C++, Java, Python, JavaScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming
- **Frontend:** React, HTML, CSS, Vite
- **Backend:** Spring Boot, FastAPI, REST APIs, WebSocket, JWT Authentication
- **Databases:** MySQL, PostgreSQL, SQLite
- **Tools & Frameworks:** Git, JavaFX/FXML, SDL2, Apache PDFBox
- **Security & CTF:** Web Exploitation, Cryptography, Reverse Engineering, Digital Forensics

Education

- **University of Dhaka** — B.Sc. in Computer Science & Engineering 2024 – Present (Expected 2028)
Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Database Management Systems, Object Oriented Design and Programming, Microcontrollers and Embedded Systems.

Technical Projects

VitalsCare — AI-Powered Community Health Platform | [GitHub](#) | *React, TypeScript, Vite, FastAPI, Python, XGBoost, Gemini LLM*

- Built an AI-powered HealthTech platform for non-communicable disease screening (Hypertension, Diabetes, CVD, Malnutrition) targeting rural and underserved communities.
- Developed disease-specific XGBoost models trained on 170k+ clinical records, achieving test accuracies of **98.3% (Hypertension)**, **97.3% (Diabetes)**, **99.8% (Malnutrition)**, and **74.5% (CVD)**.
- Implemented a Knowledge Graph engine for comorbidity reasoning, a bilingual (English/Bengali) UI, and dynamic risk dashboards with PDF report generation.

PoishaGo — Mobile Financial Service Platform | [GitHub](#) | *React, FastAPI, PostgreSQL, WebSockets, JWT, Supabase*

- Engineered a full-stack Mobile Financial Service (MFS) platform supporting P2P transfers, merchant payments, agent banking, and 2FA transaction authorization (PIN + OTP).
- Developed a real-time rule-based fraud detection engine with 5 configurable security rules, 4-role RBAC, and tiered loyalty rewards.
- Integrated WebSocket-driven live support chat and synchronized real-time transaction updates with PostgreSQL persistence.

MediMart — Full-Stack Web Application | [Frontend](#) | [Backend](#) | *React, Spring Boot, WebSocket, JWT, MySQL*

- Built a full-stack pharmacy management platform with JWT-based authentication and role-based access control for admin and customer roles.
- Implemented real-time inventory synchronization using WebSockets, enabling live stock updates across the admin dashboard.
- Designed automated PDF invoice generation and an order tracking system with admin controls for inventory management.

Escape Room Conquest — 2D Puzzle Adventure Game | [GitHub](#) | *C++, SDL2*

- Developed a multi-level escape room game in C++ using SDL2, featuring themed floors and modular puzzle systems.
- Implemented algorithmic challenges including RSA decryption, circuit puzzles, and Tetris mechanics.
- Built a contextual AI hint system (J.A.R.V.I.S.) that dynamically assists players based on in-game actions.

DSA Practice Repository | [GitHub](#) | *C++*

- Maintained a curated repository of optimized solutions covering dynamic programming, graph algorithms, trees, and number theory.
- Practiced across Codeforces, CodeChef, LeetCode, and CSES, improving algorithmic problem-solving speed and complexity optimization.

Achievements

• Hackathons & Competitions

- **Finalist**, *The Infinity AI Buildfest 2026* — Brac University, Team **DU_AlgoArchitects** (Project: **VitalsCare**, AI-powered community health risk radar platform).
- **Finalist**, *IUT Techathon 2026* — Islamic University of Technology, Team **DU_AlgoArchitects** (Project: **Vantage Robotics**, 3D digital twin & simulation suite for a 6-DOF robotic arm).
- **12th Place**, *Insight 2.0 Datathon* — Institute of Applied Statistics & Data Science, University of Dhaka, Team **DU_Neural**.

• Cybersecurity (CTF Competitions)

- Ranked **21st out of 702 teams** in *Ramadan CTF 2026* as part of **DU_CyberPhantoms**, solving challenges in web exploitation, cryptography, and reverse engineering (Score: 1415).
- Ranked **52nd** in *Al Khwarizmi CTF 2025* as part of **DU_CyberPhantoms**, placing in web exploitation and cryptography tracks.

• Competitive Programming

- Solved **700+** algorithmic problems across [Codeforces](#), [CodeChef](#), [LeetCode](#), and [CSES](#).
- Codeforces Rating: **1190** | CodeChef Rating: **1369**

Interests

Problem Solving, Application Development, Cybersecurity & CTF, Distributed Systems, Cricket